Derivation of the Inverse-Cube Force Term (k/r^3)

Binet Equation, Precession Rate, Calibration to Mercury, and Comparison to GR

> **USER PROMPT:** *Can you show the complete derivation for the effect of Inverse-Cube Force Term (k/r^3) where k is chosen to satisfy the mercury perihelion precession, and copy into docx file?*

This document presents the exact mathematical derivation of orbital motion under an inverse-square gravitational force modified by an attractive inverse-cube perturbing term ($-k/r^3$ per unit mass). We solve the differential equation of the orbit exactly, calibrate the parameter k to match Mercury's observed 43"/century anomalous perihelion precession, and analyze the resulting scaling properties.

1. Potential and Binet's Orbit Equation

Consider a central gravitational force per unit mass composed of Newton's inverse-square term and an inverse-cube perturbing term:

$$f(r) = - GM / r^2 - k / r^3$$

The corresponding effective potential per unit mass is:

$$V_{eff}(r) = - GM / r - k / (2 r^2) + h^2 / (2 r^2) = - GM / r + (h^2 - k) / (2 r^2)$$

Notice that an inverse-cube force term directly modifies the centrifugal potential barrier ($h^2/2r^2$). Binet's equation for central orbital motion in terms of $u = 1/r$ is:

$$d^2u/d\vartheta^2 + u = - (1 / (h^2 u^2)) * f(1/u)$$

Substituting $f(1/u) = - GM u^2 - k u^3$ into Binet's equation yields:

$$d^2u/d\vartheta^2 + u = (1 / (h^2 u^2)) * [GM u^2 + k u^3] = (GM / h^2) + (k / h^2) * u$$

Rearranging terms into standard harmonic oscillator form:

$$d^2u/d\vartheta^2 + (1 - k/h^2) u = GM / h^2$$

2. Exact Solution and Apsidal Precession Formula

Let $\gamma^2 = 1 - k/h^2$. For a weak perturbation ($k \ll h^2$), γ is real and positive:

$$\gamma = \sqrt{(1 - k/h^2)} \approx 1 - k / (2 h^2)$$

The general exact solution to the differential equation is:

$$u(\vartheta) = GM / (h^2 \gamma^2) * [1 + e \cos(\gamma \vartheta)]$$

Pericenter occurs whenever the cosine term equals +1, which requires $\gamma \theta = 2\pi m$ for integers m. The angular change between consecutive pericenter passages is:

$$\Delta\vartheta = 2\pi / \gamma = 2\pi / \sqrt{(1 - k/h^2)} \approx 2\pi * [\ 1 + k / (2\ h^2)\ ]$$

The perihelion precession advance per orbit ($\Delta\varpi_orbit = \Delta\theta - 2\pi$) is:

$$\Delta\varpi_orbit = \pi * (k / h^2) \quad \text{radians / orbit}$$

Substituting the specific angular momentum $h^2 = GM\ a\ (1 - e^2)$, where a is the semi-major axis and e is the orbital eccentricity:

$$\Delta\varpi_orbit = (\pi\ k) / [\ GM\ a\ (1 - e^2)\ ] \quad \text{radians / orbit}$$

3. Calibration of k Using Mercury's Perihelion Precession

Mercury's observed anomalous perihelion precession is 43" per century. Mercury completes $N_Merc \approx 415.2$ revolutions per century (orbital period $P \approx 0.24084$ years).

- **Advance per orbit:** 43" / 415.2 orbits ≈ 0.10356" per orbit
- **Radian conversion:** 0.10356" × (π / 648,000 rad/") ≈ 5.021×10^{-7} rad/orbit

Equating $\Delta\varpi_orbit = \pi\ (k / h_Merc^2)$ to 5.021×10^{-7} rad/orbit allows us to determine the dimensionless ratio k / h_Merc^2:

$$k / h_Merc^2 = \Delta\varpi_orbit / \pi = (5.021 \times 10^{-7}) / \pi \approx 1.598 \times 10^{-7}$$

To compute the physical value of k, we use Mercury's specific angular momentum squared, $h_Merc^2 = GM_\odot\ a_Merc\ (1 - e_Merc^2)$:

- **Solar mass parameter:** $GM_\odot \approx 1.327 \times 10^{20}\ m^3/s^2$
- **Mercury semi-major axis:** $a_Merc \approx 5.791 \times 10^{10}$ m
- **Mercury eccentricity:** $e_Merc \approx 0.2056 \Rightarrow 1 - e_Merc^2 \approx 0.9577$

$$h_Merc^2 \approx (1.327 \times 10^{20}) * (5.791 \times 10^{10}) * (0.9577) \approx 7.36 \times 10^{30}\ m^4/s^2$$

Multiplying by the dimensionless ratio:

$$k = (1.598 \times 10^{-7}) * (7.36 \times 10^{30}\ m^4/s^2) \approx 1.176 \times 10^{24}\ m^4/s^2$$

4. Scaling Properties and Comparison with General Relativity

The inverse-cube perturbing force exhibits notable scaling characteristics across different orbits:

- **Distance Dependence:** Unlike Hall's hypothesis ($F \propto 1/r^{(2+\delta)}$), where $\Delta\varpi_orbit = \pi\delta$ was identical for all orbits regardless of size, the inverse-cube force produces an advance per orbit that scales inversely with semi-latus rectum $p = a(1 - e^2)$: $\Delta\varpi_orbit \propto 1 / [\ a(1 - e^2)\ ]$.
- **Comparison to GR:** In General Relativity, Schwarzschild geometry introduces an extra term in Binet's equation proportional to u^2 (corresponding to a $1/r^4$ force perturbation). The relativistic pericenter advance is $\Delta\varpi_GR = (6\pi\ GM) / [\ c^2\ a\ (1 - e^2)\ ]$.
- **Radial Scaling Agreement:** Consequently, calibrating an inverse-cube force term to Mercury predicts the exact same $1/[a(1 - e^2)]$ radial scaling across planetary orbits as General Relativity.

Analysis of Inverse-Cube Force Term (k/r^3) Violations

> **USER PROMPT:** *Does the inverse cube term create any violations of keplers third law, or other planet precessions, or lunar motion, assuming k is chosen to satisfy the mercury perihelion precession, and violations based on pre-1900 physics?*

Assuming the parameter k in the inverse-cube force perturbation $f(r) = -GM/r^2 - k/r^3$ is calibrated to match Mercury's anomalous 43"/century perihelion precession ($k \approx 1.176 \times 10^{24}$ m^4/s^2), we evaluate whether this modified force law introduces detectable violations in Kepler's Third Law, other planetary precessions, or lunar orbital motion under 19th-century observational limits.

1. Kepler's Third Law Distortion

For a circular orbit of radius a, the modified radial acceleration requirement gives:

$$n^2 a = GM / a^2 + k / a^3 \quad \Rightarrow \quad n^2 a^3 = GM * [\ 1 + k / (GM\ a)\ ]$$

The standard Keplerian constant $n^2 a^3$ is modified by a tiny distance-dependent fractional shift $\varepsilon(a) = k / (GM\ a)$:

- **Mercury (a = 0.387 AU):** $\varepsilon(a_{Merc}) = (1.176 \times 10^{24}) / [\ (1.327 \times 10^{20}) \times (5.791 \times 10^{10})\ ] \approx 1.53 \times 10^{-7}$
- **Earth (a = 1.000 AU):** $\varepsilon(a_{Earth}) = (1.176 \times 10^{24}) / [\ (1.327 \times 10^{20}) \times (1.496 \times 10^{11})\ ] \approx 0.59 \times 10^{-7}$

Conclusion: The differential shift in $n^2 a^3$ across the solar system is on the order of $\sim 10^{-7}$. In pre-1900 astronomical observations, solar mass parameters and orbital semi-major axes were known to an accuracy of $\sim 10^{-4}$ to 10^{-5}. Consequently, this shift was completely unobservable and created NO effective violation of Kepler's Third Law.

2. Precession of Other Planets (Venus, Earth, Mars)

The per-orbit precession advance under an inverse-cube force is $\Delta\varpi_orbit = (\pi\ k) / [\ GM\ a\ (1 - e^2)\ ]$. Multiplying by orbital frequency $N_p = 1 / P \propto a^{-3/2}$ yields the annual or secular precession rate:

$$\dot{\varpi}_p = N_p * \Delta\varpi_orbit \propto 1 / [\ a\text{^}(5/2)\ (1 - e^2)\ ]$$

Crucially, this radial scaling $1 / [\ a\text{^}(5/2)\ (1 - e^2)\ ]$ is mathematically IDENTICAL to the precession rate predicted by General Relativity. Evaluating for inner planets:

- **Venus (a = 0.723 AU, e = 0.0068):** Predicted: 8.6" / century (Observed pre-1900 residual: ~16.8" ± 10")
- **Earth (a = 1.000 AU, e = 0.0167):** Predicted: 3.8" / century (Observed pre-1900 residual: ~3.7" ± 1")
- **Mars (a = 1.524 AU, e = 0.0934):** Predicted: 1.35" / century (Observed pre-1900 residual: ~5.0" ± 3")

Conclusion: Because the inverse-cube force scales identically to General Relativity across planetary semi-major axes, it produces NO unexpected discrepancies for other planets and matches pre-1900 observational residuals just as well as GR.

3. Lunar Motion (The Earth-Moon System)

While planetary orbits remain consistent, applying the Mercury-calibrated inverse-cube force to the Earth-Moon system leads to a severe observational breakdown.

Scenario A: Universal Constant k (k independent of central mass M)

If k is a universal constant ($k = 1.176 \times 10^{24}$ m⁴/s²), substituting Earth's gravitational parameter $GM_{\oplus} \approx 3.986 \times 10^{14}$ m³/s² and Moon's orbit ($a = 3.844 \times 10^{8}$ m) yields:

$$\Delta\varpi_{Moon} = (\pi k) / [GM_{\oplus} a_{Moon} (1 - e_{Moon}^2)] \approx 24.2 \text{ radians / orbit}$$

This predicts that the Moon's perigee precesses by multiple full revolutions per single orbit (~1385°/orbit)—a catastrophic violation that immediately rules out a universal k.

Scenario B: Mass-Proportional Coupling $k = \alpha GM$ ($\alpha \approx 8862$ m²)

If k scales with central mass M (like gravity), the per-orbit advance becomes $\Delta\varpi_{orbit} = (\pi \alpha) / [a (1 - e^2)]$.

- **Per-orbit advance:** $\Delta\varpi_{Moon, orbit} = [\pi \times (8862 \text{ m}^2)] / [3.844 \times 10^{8} \text{ m} \times 0.997] \approx 7.26 \times 10^{-5}$ rad/orbit ≈ 15.0" / orbit
- **Secular precession rate:** $\dot{\varpi}_{Moon} = 1336.8$ orbits/century × 15.0" / orbit ≈ 20,000" / century (≈ 5.5° / century)

Conclusion: Pre-1900 lunar observational theories (Delaunay, Hansen, Newcomb) constrained unmodeled lunar perigee drift to under ~2" per century. An extra precession of ~20,000"/century over-predicts the observed limit by a factor of nearly 10,000. This represents a FATAL VIOLATION that conclusively ruled out an inverse-cube force in 19th-century celestial mechanics.

Parametrization $k = \alpha(GM)^2$: Relativistic Connection & Lunar Resolution

Dimensional Analysis, Equivalence to General Relativity, and Suppression of Lunar Drift

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> **USER PROMPT:** *what is k = alpha (GM)^2*

The parametrization $k = \alpha(GM)^2$ represents the unique scaling law where the strength of the inverse-cube term depends quadratically on the central body's gravitational mass GM. This choice reveals the direct connection to General Relativity and resolves the lunar motion anomaly.

1. Dimensional Analysis and the Speed of Light

- **Dimensions of k:** From $f(r) = -k/r^3$, the parameter k has dimensions $[k] = L^4 T^{-2}$.
- **Dimensions of $(GM)^2$:** Since $[GM] = L^3 T^{-2}$, its square has dimensions $[(GM)^2] = L^6 T^{-4}$.

Taking the ratio yields the dimensions of α:

$$dim(\alpha) = [k] / [(GM)^2] = (L^4 T^{-2}) / (L^6 T^{-4}) = T^2 / L^2 = 1 / velocity^2$$

Because α has dimensions of inverse velocity squared, fundamental physics requires $\alpha = \beta / c^2$, where c is the speed of light and β is a dimensionless constant. Thus:

$$k = \beta (GM)^2 / c^2$$

2. Exact Equivalence to General Relativity

Substituting $k = \beta(GM)^2 / c^2$ into the per-orbit precession advance formula $\Delta\varpi_orbit = (\pi k) / [GM a (1 - e^2)]$ yields:

$$\Delta\varpi_orbit = \pi [\beta (GM)^2 / c^2] / [GM a (1 - e^2)] = (\beta \pi GM) / [c^2 a (1 - e^2)]$$

Setting the dimensionless constant $\beta = 6$ (giving $\alpha = 6/c^2 \approx 6.68 \times 10^{-17}\ s^2/m^2$):

$$\Delta\varpi_orbit = (6 \pi GM) / [c^2 a (1 - e^2)]$$

This matches the exact perihelion precession formula derived from Schwarzschild geometry in General Relativity.

3. Resolution of the Lunar Precession Anomaly

Scaling k quadratically with M naturally suppresses the perturbing effect for smaller central bodies like Earth ($M_\oplus << M_\odot$):

System	Central Mass GM	Perturbation Constant k	Precession Rate ($\dot{\varpi}$)	Pre-1900 Limit
Mercury / Sun	$1.327 \times 10^{20}\ m^3/s^2$	$1.176 \times 10^{24}\ m^4/s^2$	43.0" / century	Matches 43"/century
Moon / Earth	$3.986 \times 10^{14}\ m^3/s^2$	$1.062 \times 10^{13}\ m^4/s^2$	~0.06" / century	< 2.0"/century

By replacing a linear or constant k with $k = 6(GM)^2/c^2$, the predicted lunar perigee drift drops from an unphysical 20,000"/century down to just 0.06"/century, fully satisfying pre-1900 observational constraints.